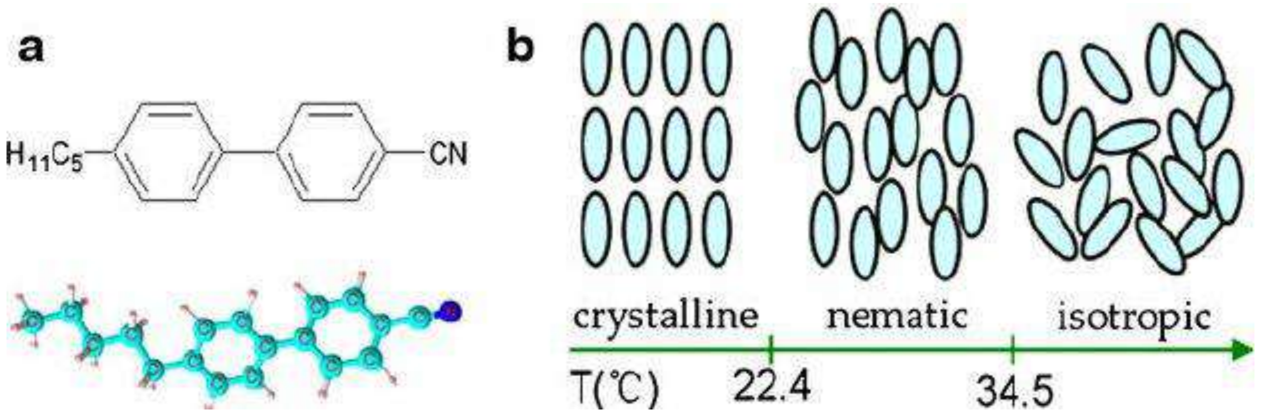
Microplastic Detection using 5CB Liquid Crystal

Data Analysis by Sharjeel Ahmad (24100083)

Research Objectives

- The research aims to see whether or not it is possible to detect micro plastics in a given sample by observing the unique kinds of optical or non-optical interactions (if any) between the liquid crystal and the plastic's surface
- The research also hopes to be able to differentiate between different types of micro-plastics between exploiting their varied chemical structures and subsequent interactions with the liquid crystal in question

5CB



5CB refers to the nematic liquid crystal shown in the diagram above, it has a chemical formula of $C_{18}H_{19}N$.

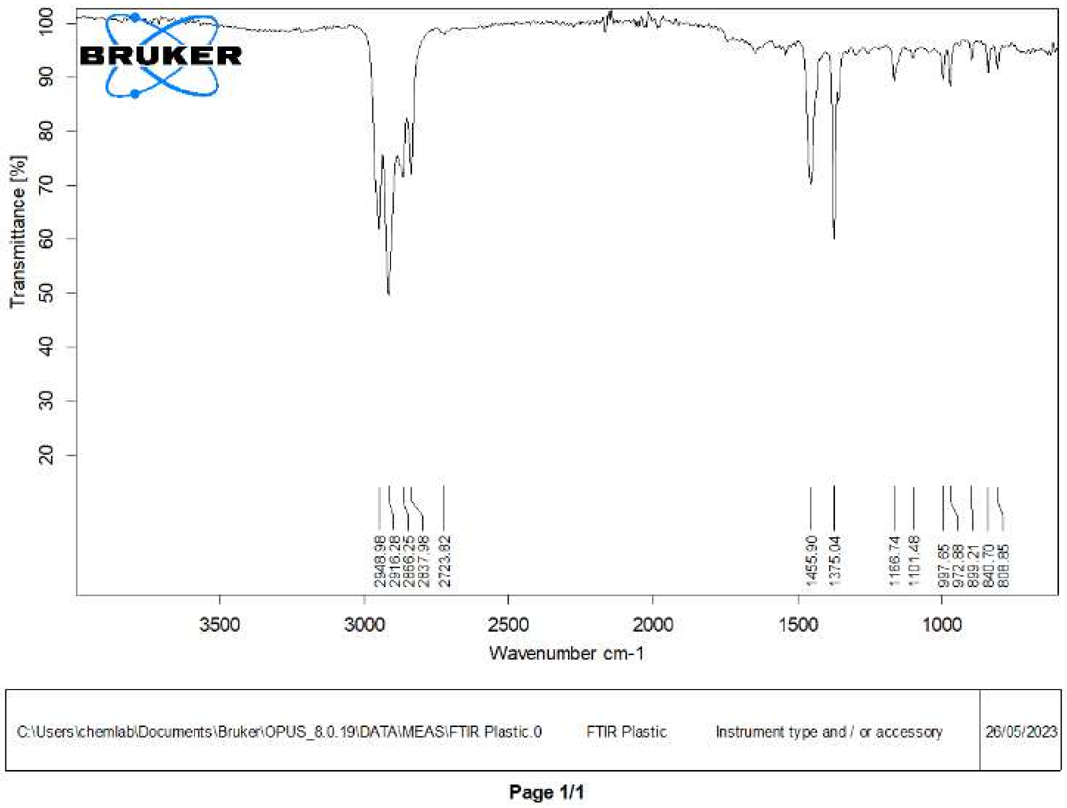
The liquid crystal 5CB undergoes a phase transition from a crystalline state to a nematic state at 22.5 °C and it goes from a nematic to an isotropic state at 35 °C.

It is birefringent and displays various optical properties based on the polarization of the light used to observe it.

In our analysis, a layer of 5CB is spin coated onto the plastic sample which is placed on a cleaned glass slide.

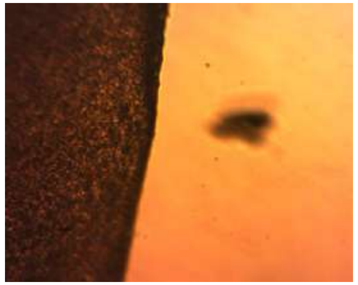
Low Density Polyethylene

FTIR results indicate the following sample has the closest match to low density polyethylene which was the initial claim as well when obtaining the plastic sample.



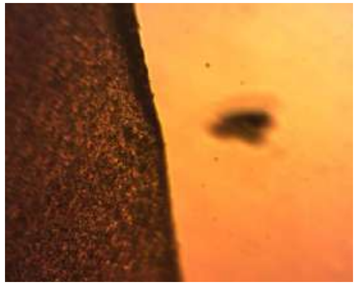
Uncrossed Polyethylene (600 μm) Birefringence Test via Rotation

0^{o}



10^{o}

20^{o}



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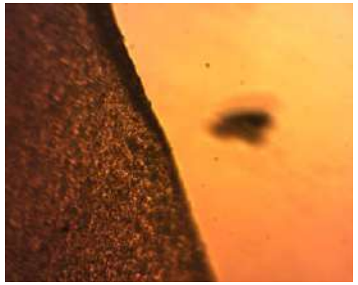

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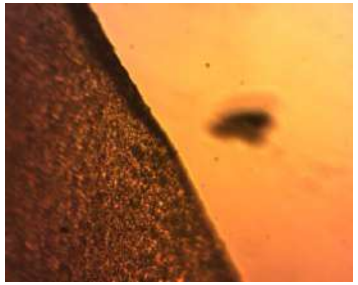
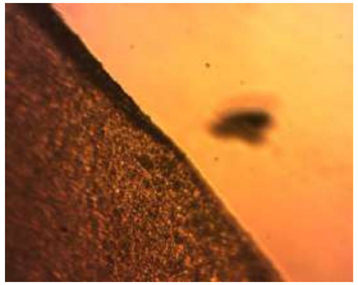
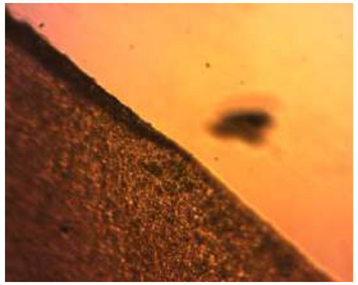


50^{o}



60^{o}



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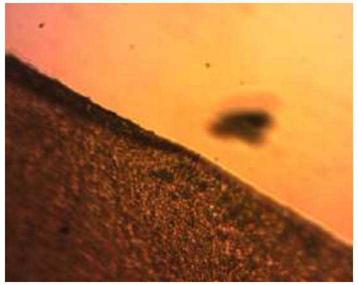
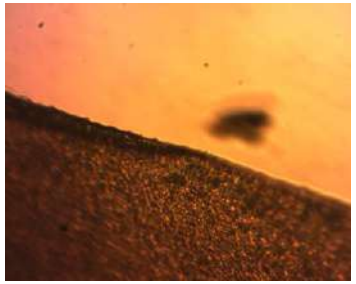
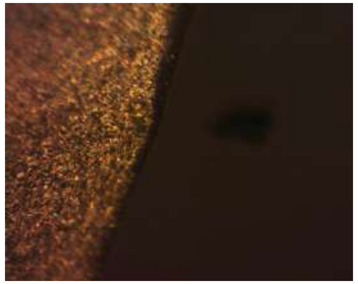


80^{o}



90^{o}

Crossed Polyethylene (600 μm) Birefringence Test via Rotation



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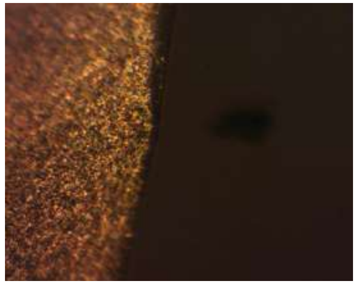

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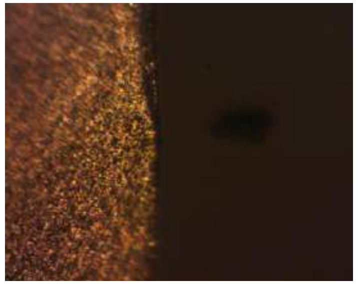

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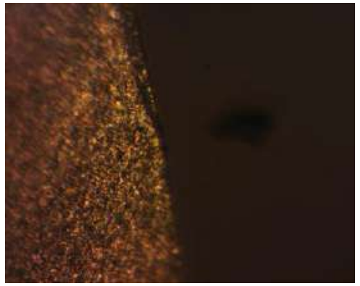

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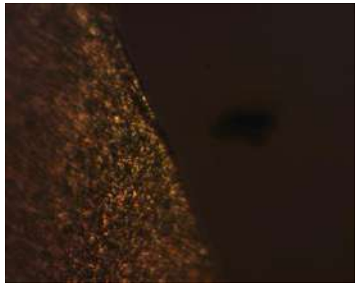

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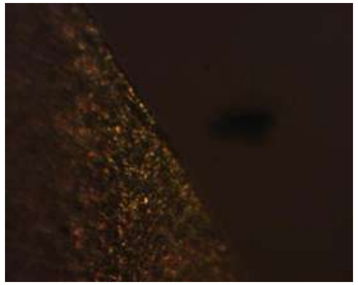

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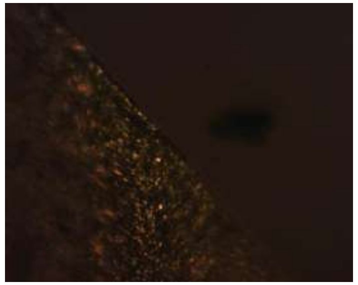

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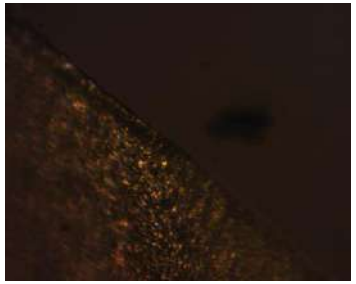

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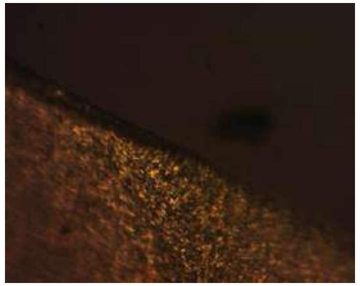

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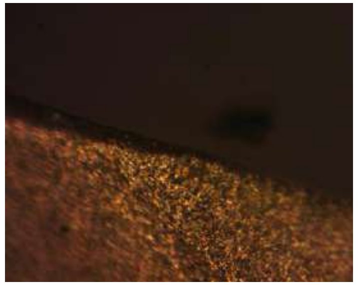
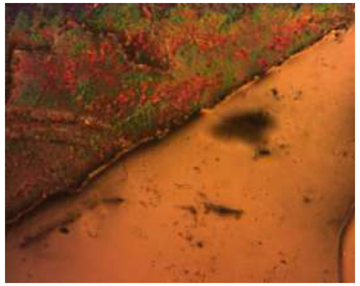


90^{o}

Uncrossed Polyethylene (600 µm) + 5CB Birefringence Test via Rotation

0^{o}



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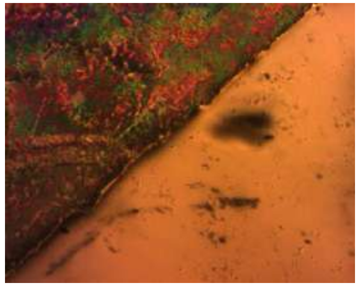

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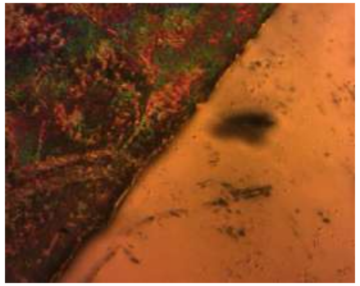

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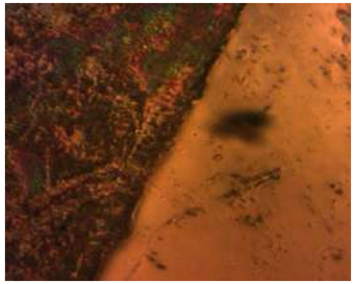

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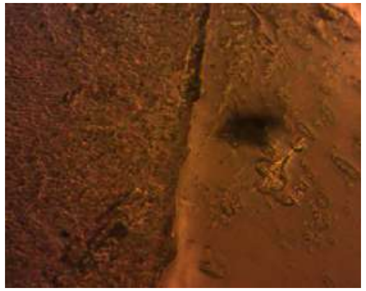
50^{o}



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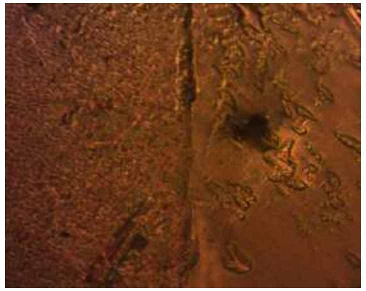

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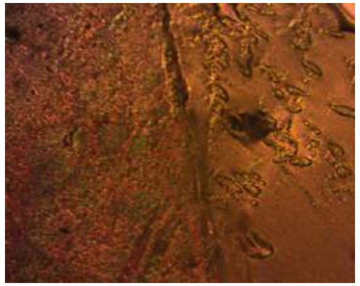

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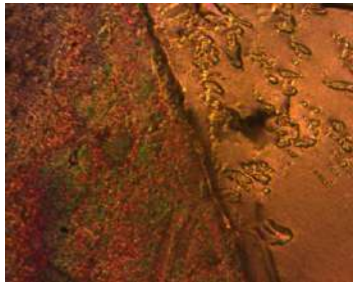
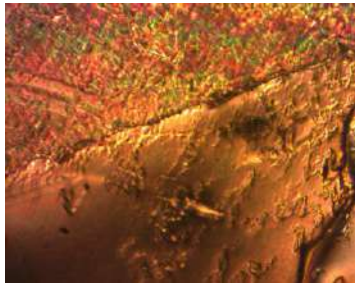


90^{o}

Crossed Polyethylene (600 μm) + 5CB Birefringence Test via Rotation



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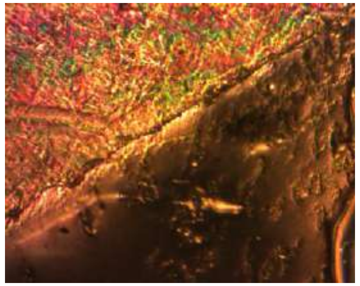
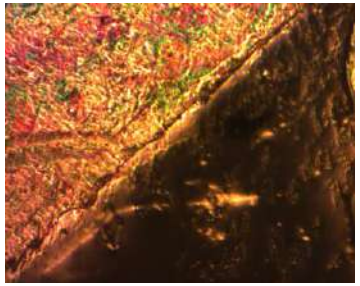


10°



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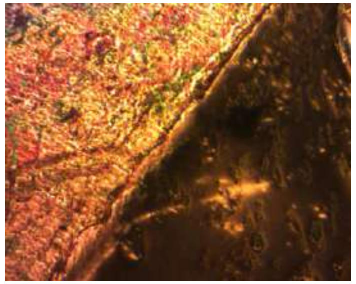

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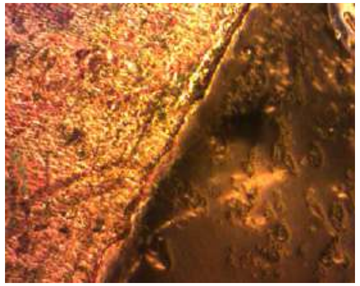

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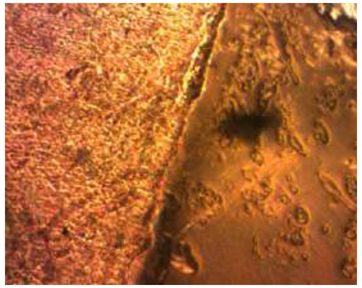

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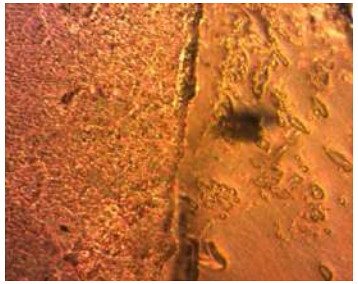

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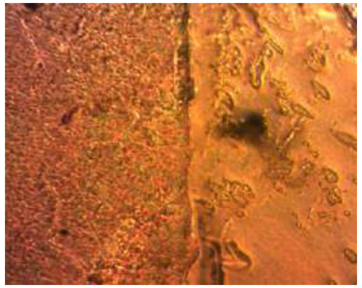

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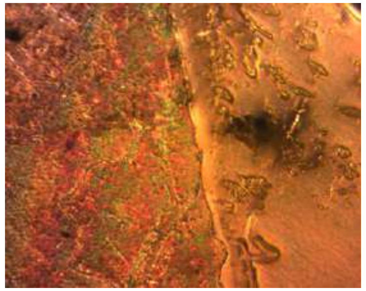

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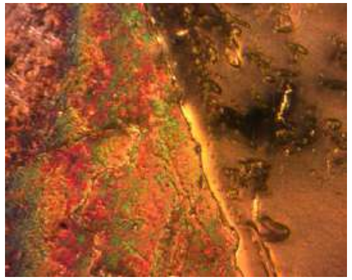
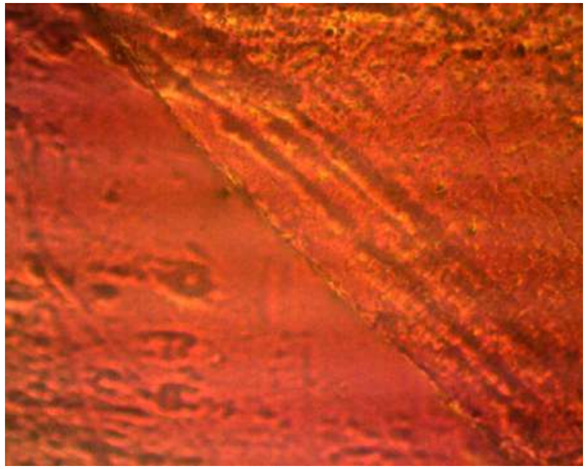
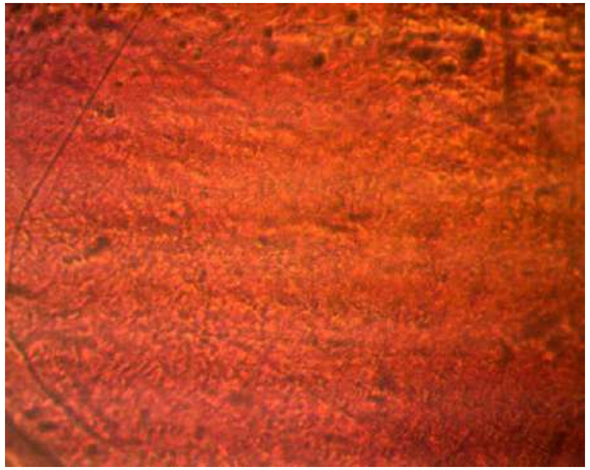


90°

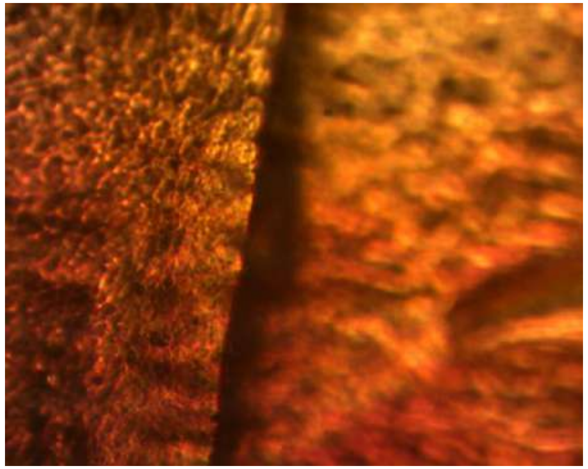
Uncrossed Polarizer Microscopic Images with 5CB and Polyethylene (600 μm) Sample



Left Boundary



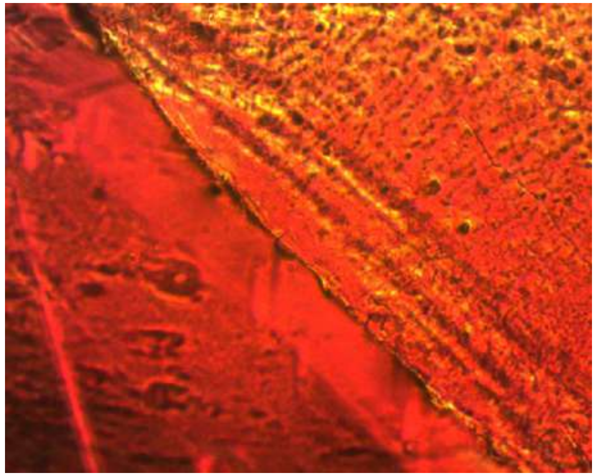
Center



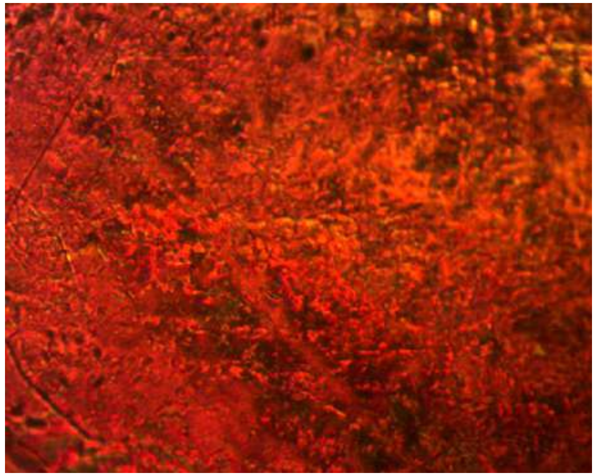
Right Boundary

Zoomed out photo taken from a different microscope

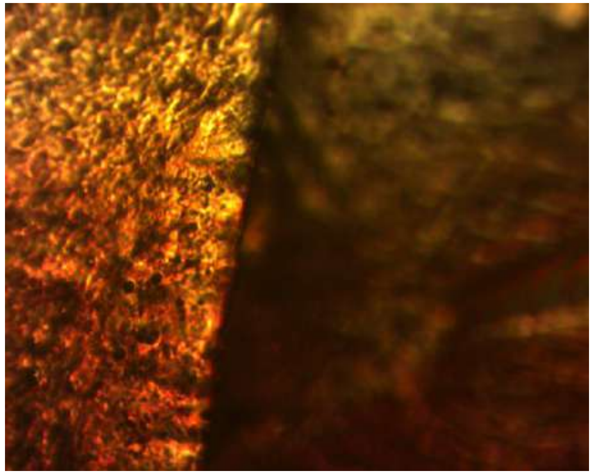
Crossed Polarizer Microscopic Images with 5CB and Polyethylene (600 μm) Sample



Left Boundary

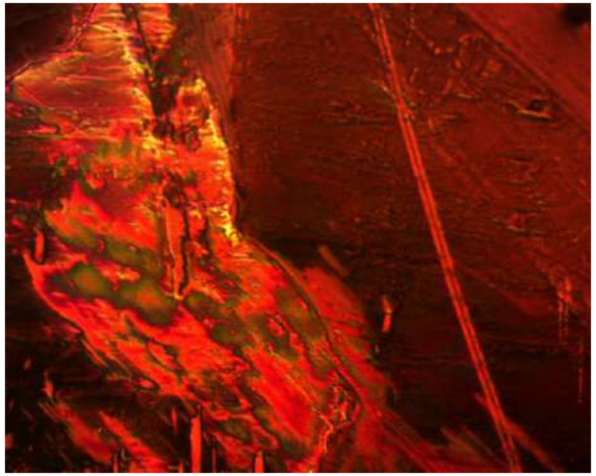


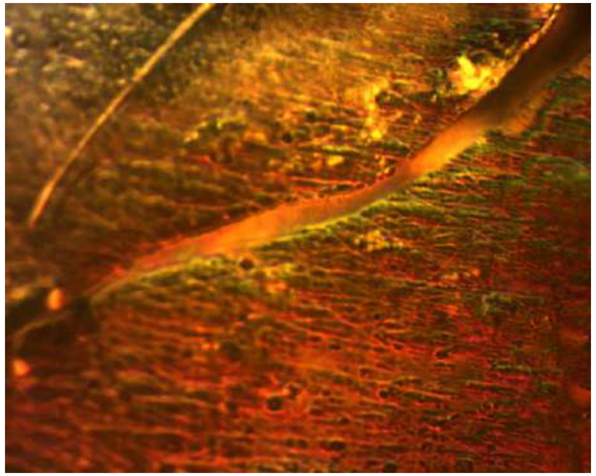
Center



Right Boundary

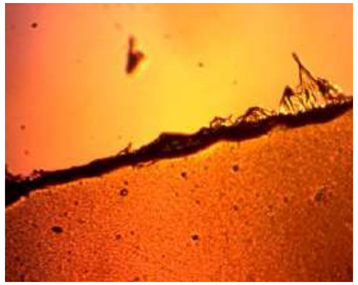
5CB on left edge (away from plastic layer)





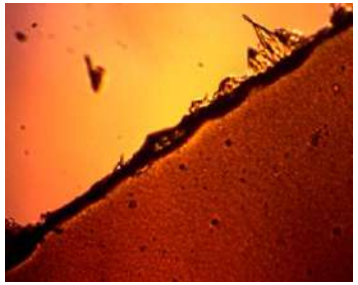
5CB on right edge (away from plastic layer)

Uncrossed Polyethylene (60 μm) Birefringence Test via Rotation

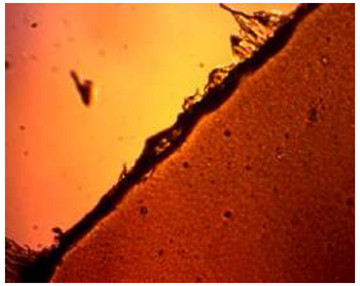


0^o

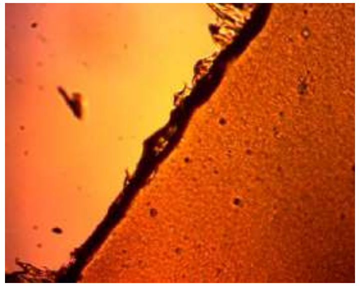
10^o



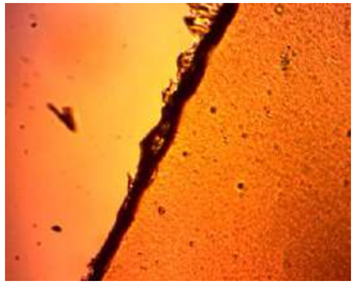
20^o



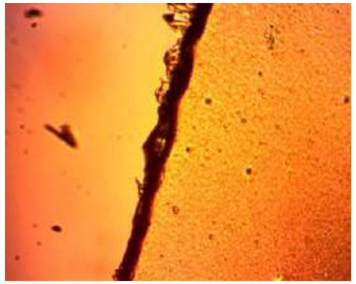
30^o



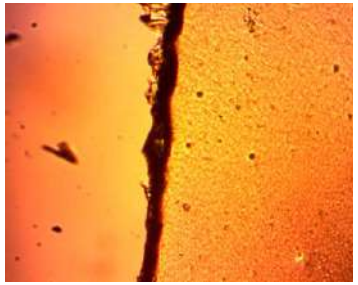
40^o



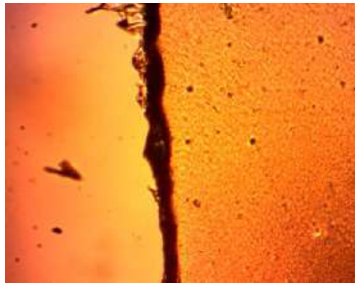
50^o



60^o



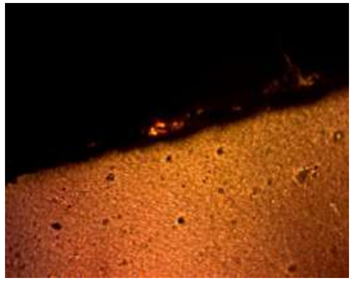
70^o



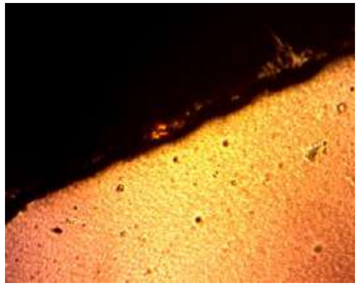
80^o

90^o

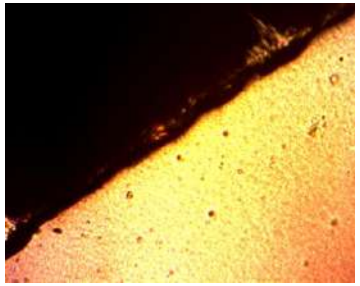
Crossed Polyethylene (60 μm) Birefringence Test via Rotation



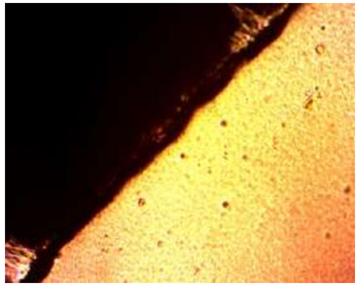
0°



10°



20°



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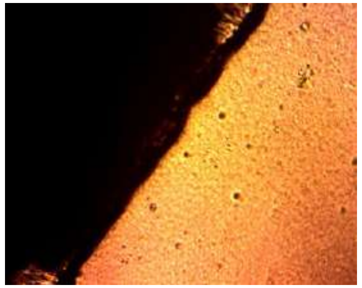

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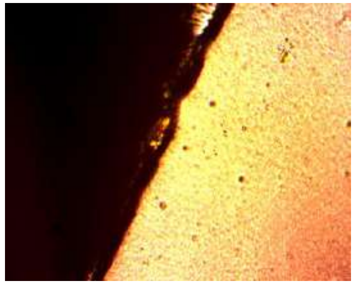
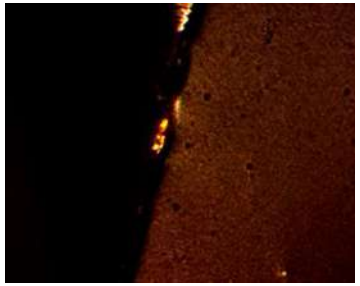
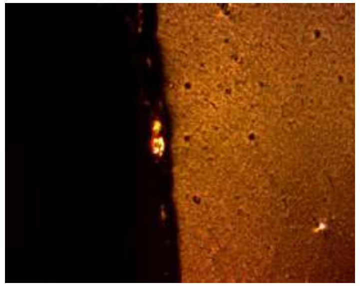


50°

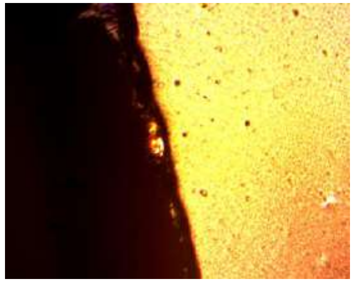


60°

70°

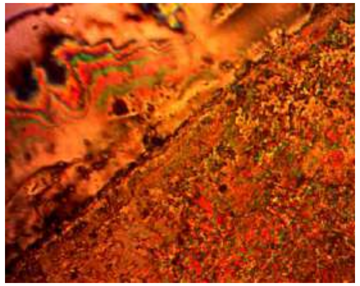


80°

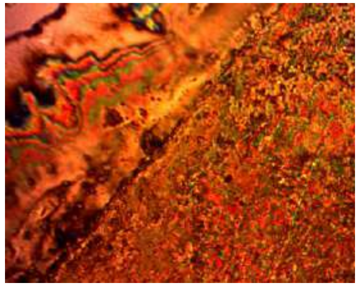


90°

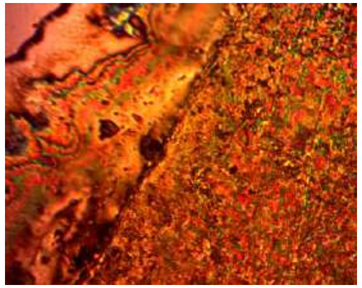
Uncrossed Polyethylene (60 μm) + 5CB Birefringence Test via Rotation



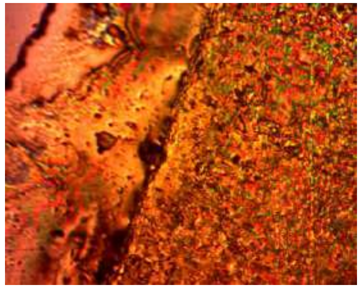
0°



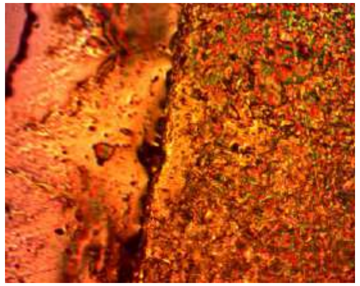
10°



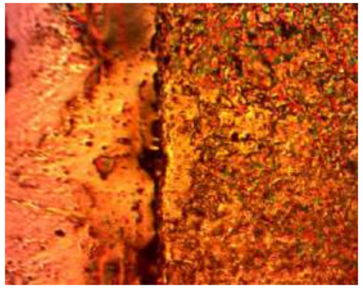
20°



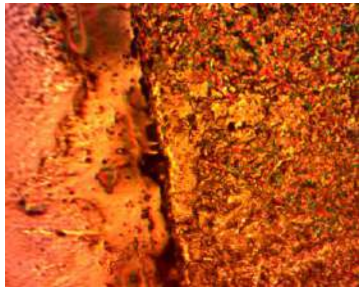
30°



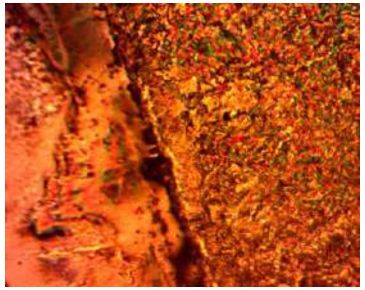
40°



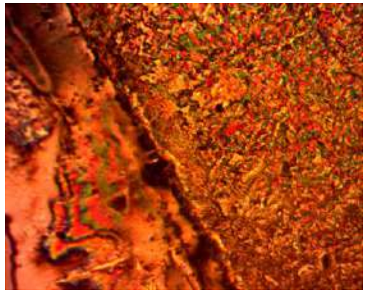
50°



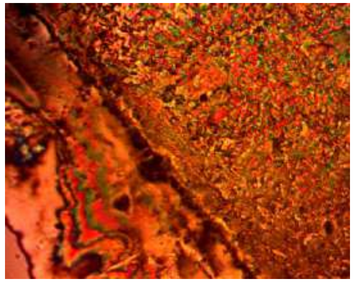
60°



70°

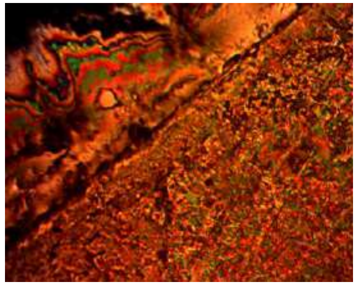


80°

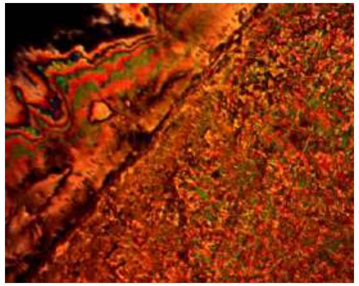


90°

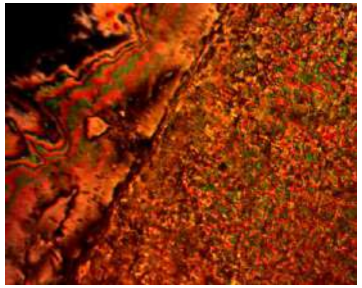
Crossed Polyethylene (60 μm) + 5CB Birefringence Test via Rotation



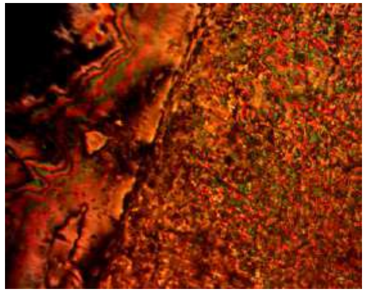
0^{o}



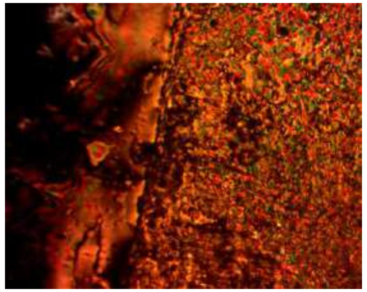
10^{o}



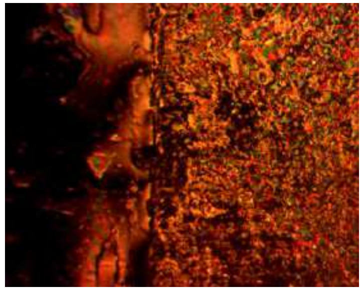
20^{o}



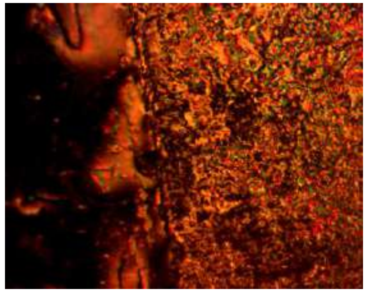
30^{o}



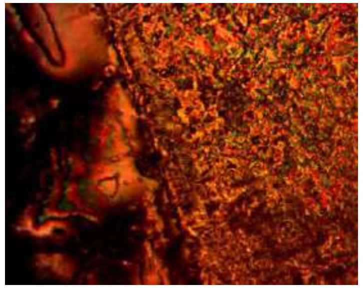
40^{o}



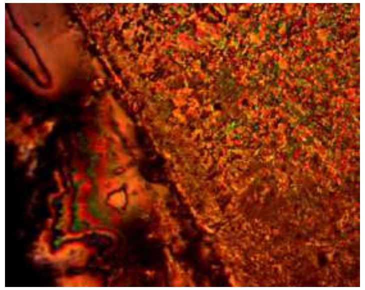
50^{o}



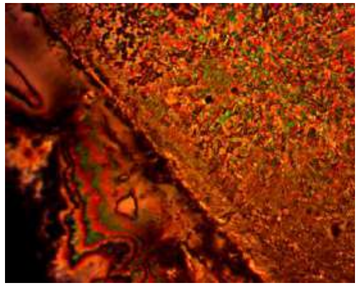
60^{o}



70^{o}



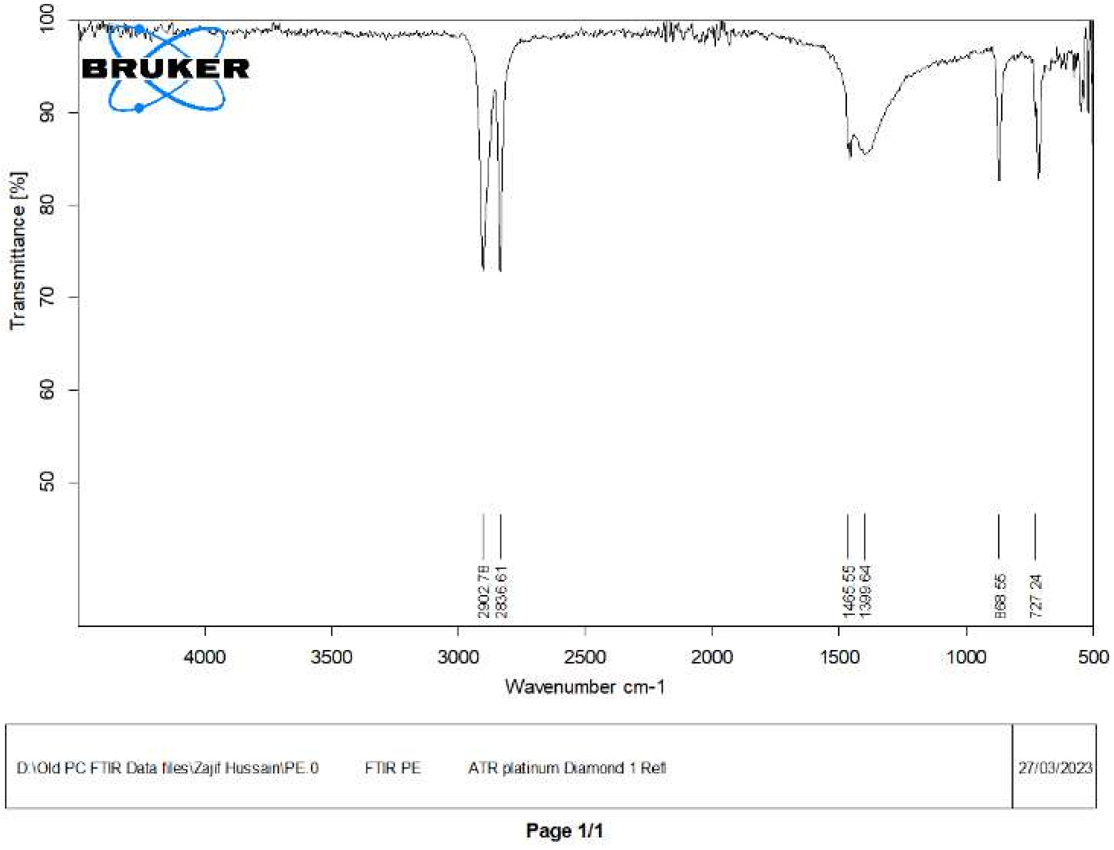
80^{o}



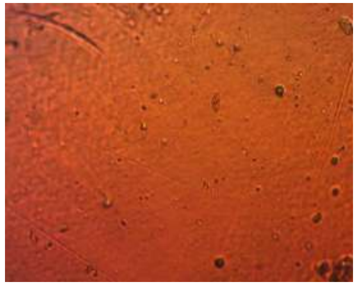
90^{o}

Low Density Polypropylene

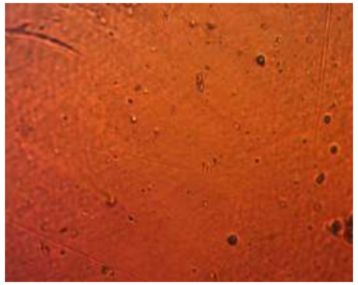
FTIR results indicate the following sample has the closest match to low density polypropylene which was the initial claim as well when obtaining the plastic sample.



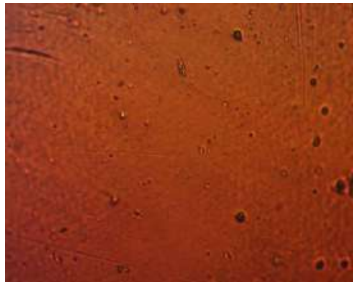
Polypropylene Birefringence (100 µm) Test via Rotation



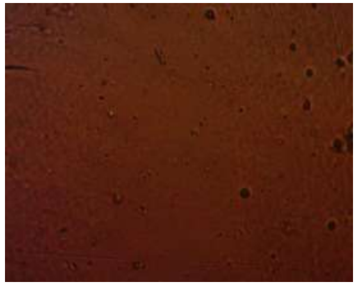
0^o



10^o



20^o

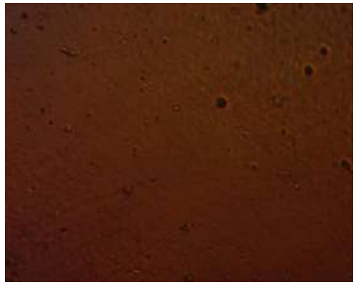


30^o

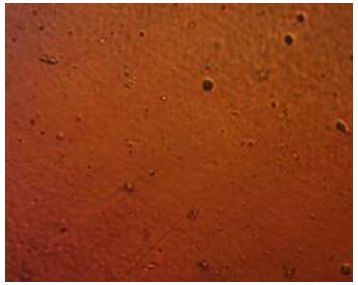
40^o

50^o

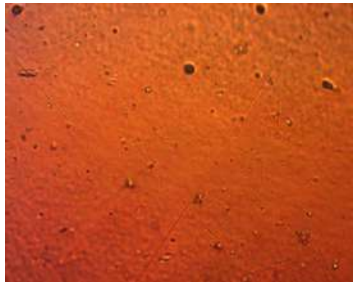
60^o



70^o

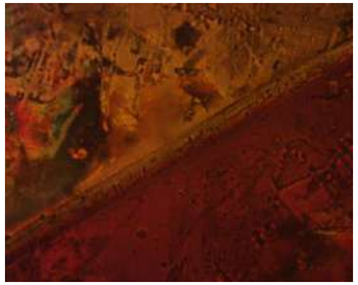


80^o



90^o

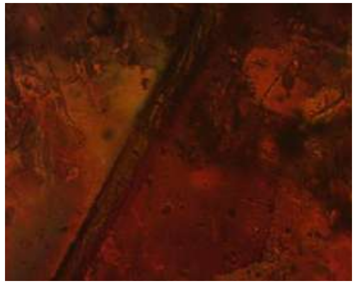
Polypropylene (100 μm) + 5CB Birefringence Test via Rotation



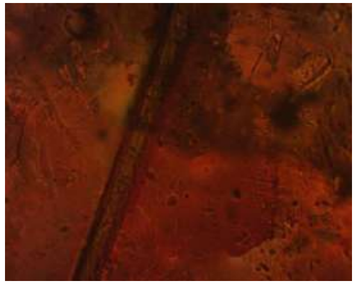
0°

10°

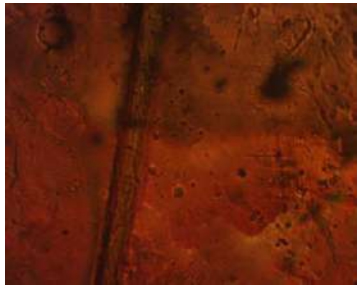
20°



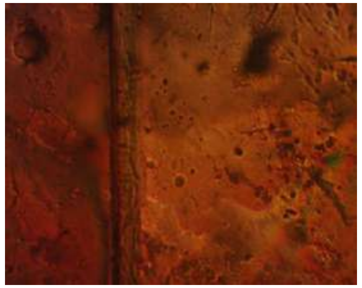
30°



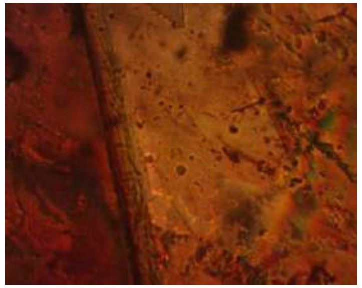
40°



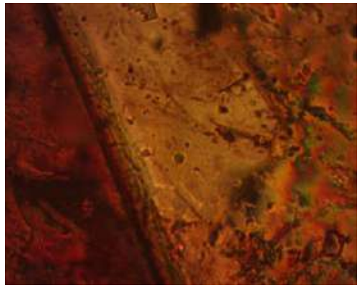
50°



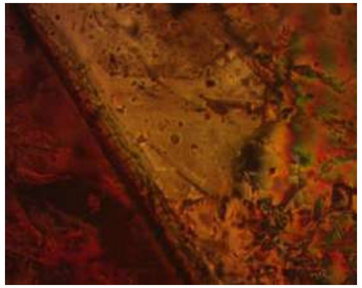
60°



70°

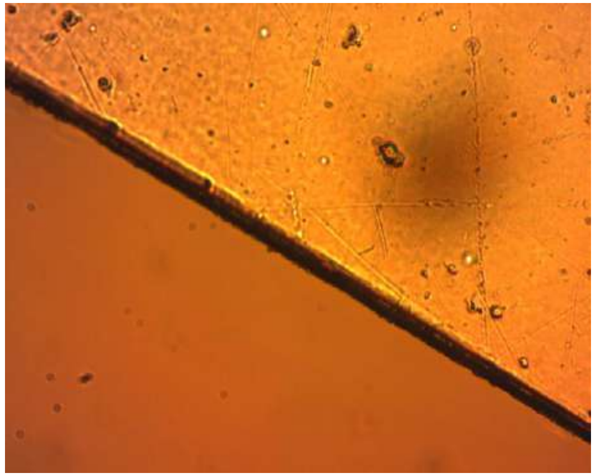


80°

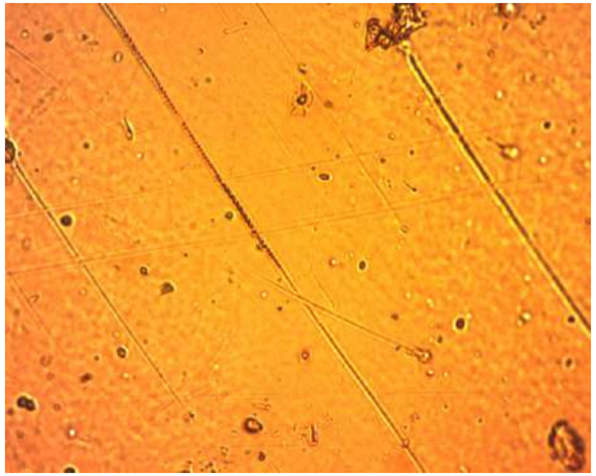


90°

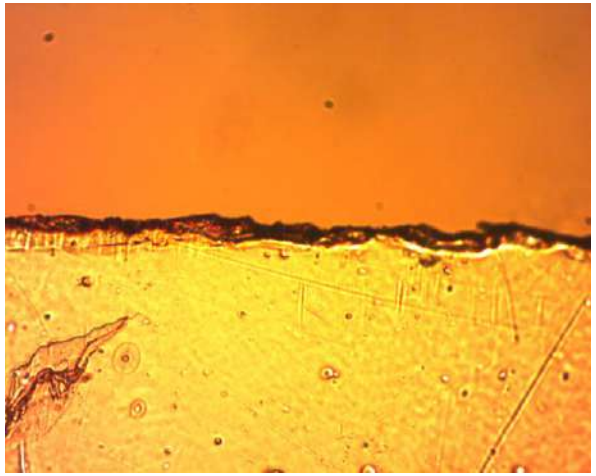
Uncrossed Polarizer Microscopic Images with Polypropylene Sample



Bottom Boundary



Center

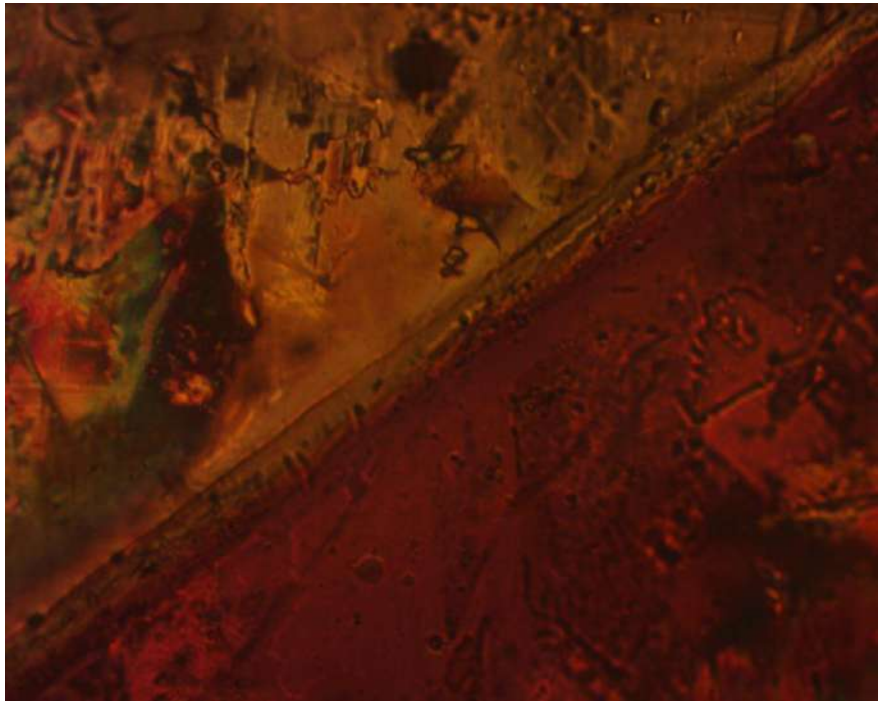


Top Boundary

Uncrossed Polarizer Microscopic Images with Polypropylene Sample + 5CB

Side Edge

Crossed Polarizer Microscopic Images with Polypropylene Sample + 5CB



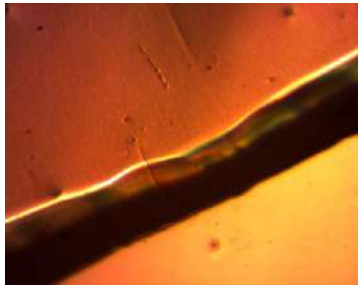
Side Edge

Zoomed Microscopic Images with Polypropylene Sample + 5CB

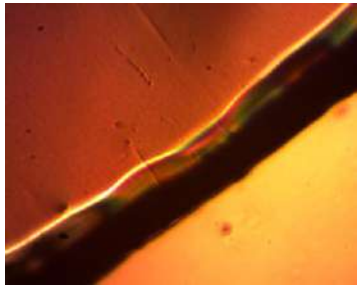
Side Edge

Low Density Polyethylene Terephthalate

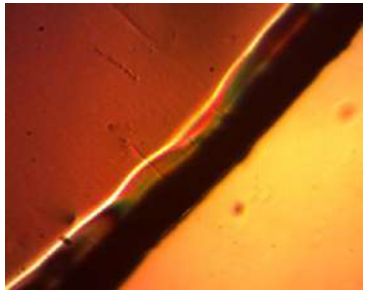
Uncrossed PET Aquafina (67 μm) Birefringence Test via Rotation



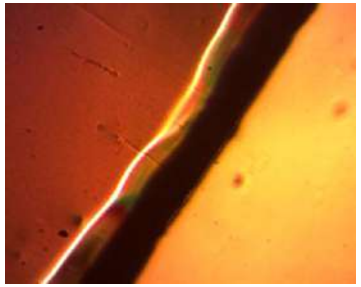
0°



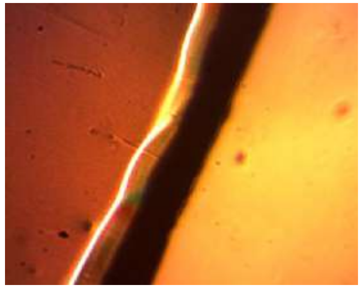
10°



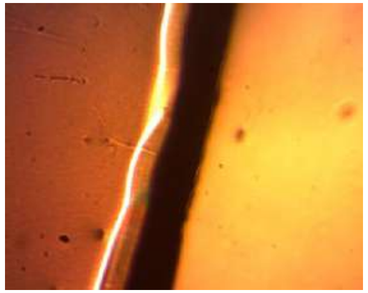
20°



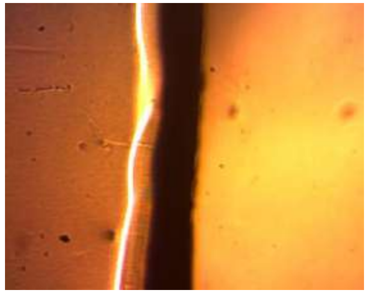
30°



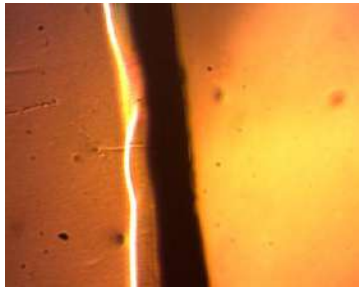
40°



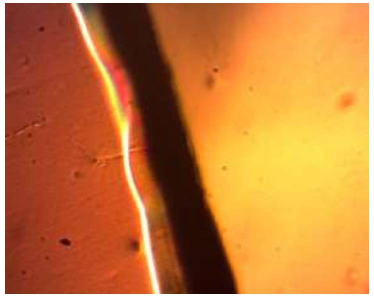
50°



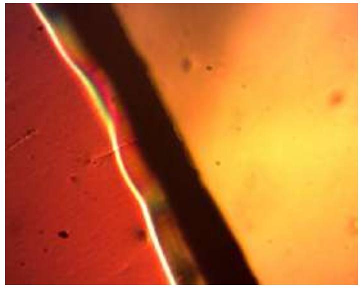
60°



70°

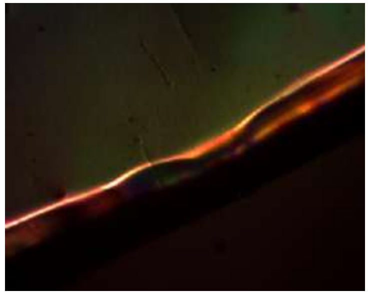


80°

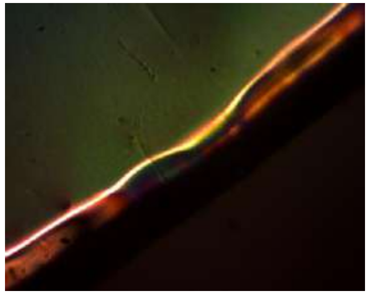


90°

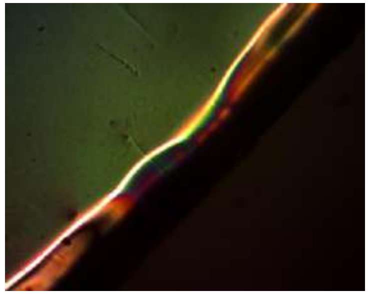
Crossed PET Aquafina (67 μm) Birefringence Test via Rotation



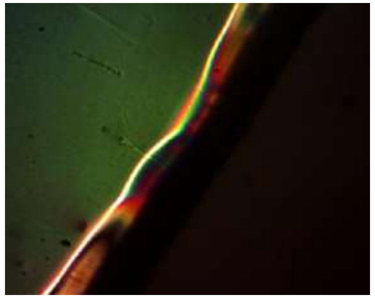
0^{o}



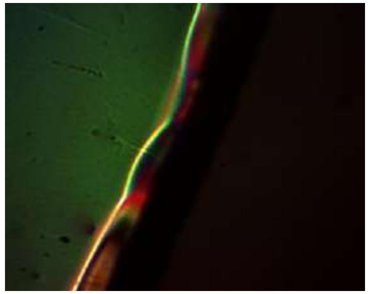
10^{o}



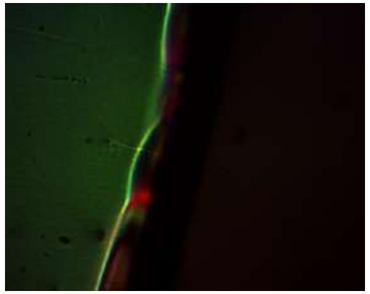
20^{o}



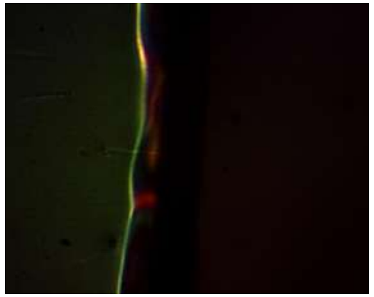
30^{o}



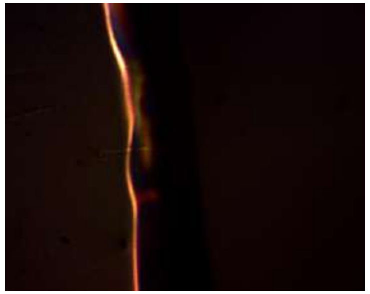
40^{o}



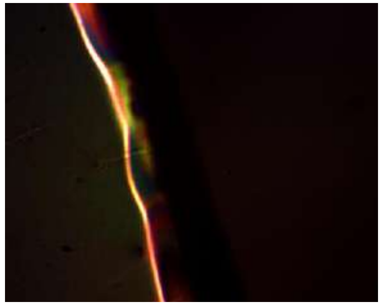
50^{o}



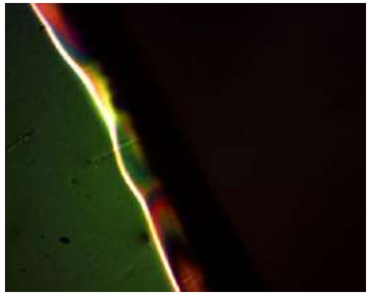
60^{o}



70^{o}

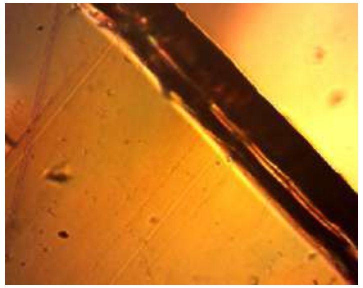


80^{o}

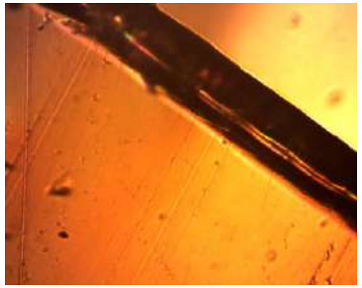


90^{o}

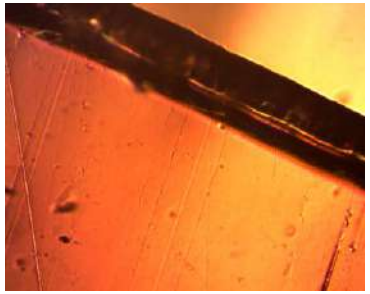
Uncrossed PET Nestle (67 μm) Birefringence Test via Rotation



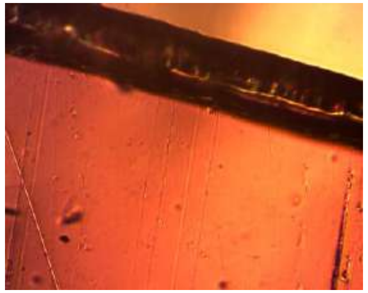
0^{o}



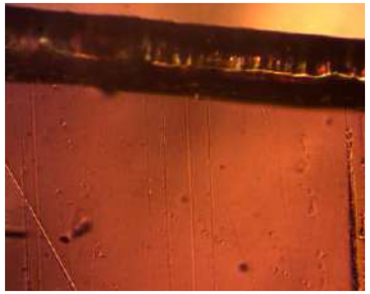
10^{o}



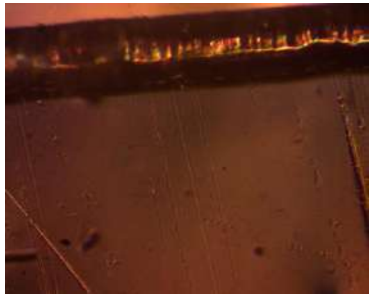
20^{o}



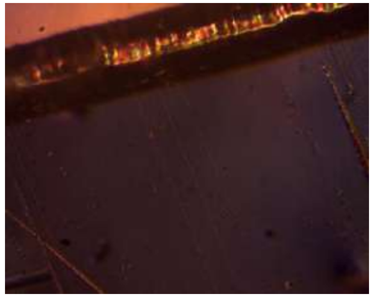
30^{o}



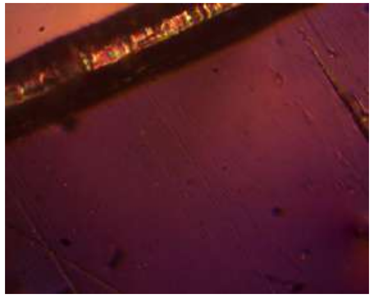
40^{o}



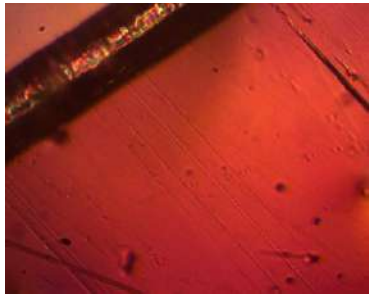
50^{o}



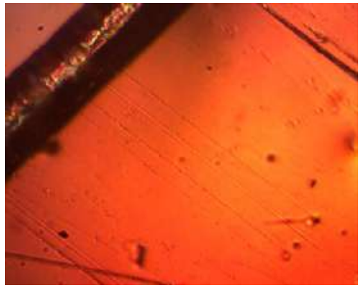
60^{o}



70^{o}

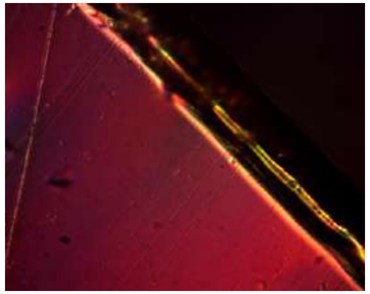


80^{o}

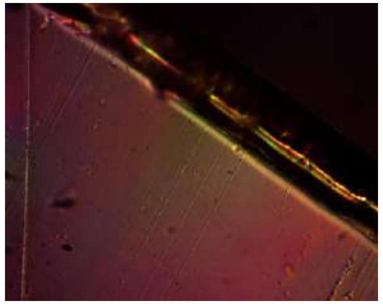


90^{o}

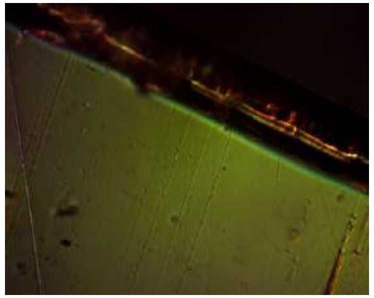
Crossed PET Nestle (67 μm) Birefringence Test via Rotation



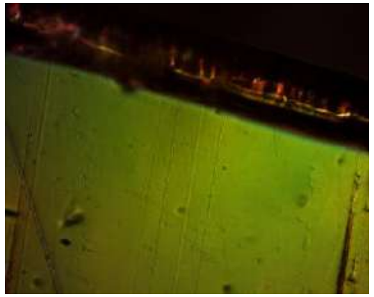
0°



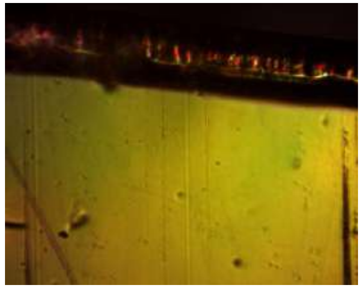
10°



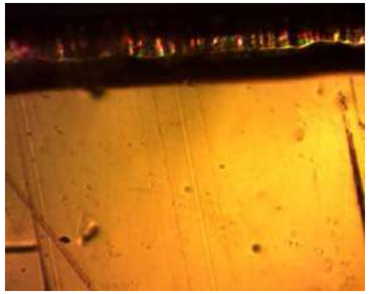
20°



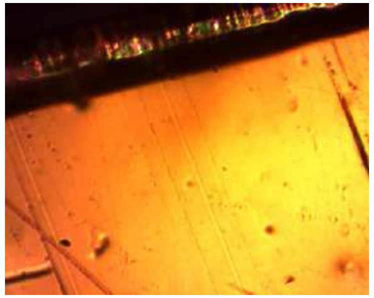
30°



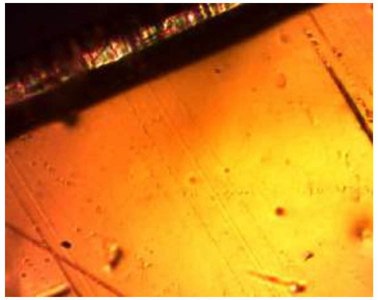
40°



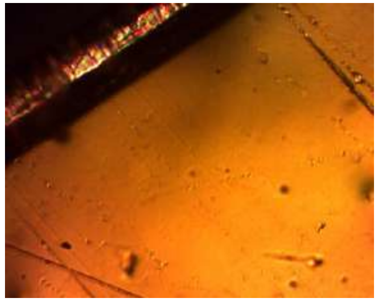
50°



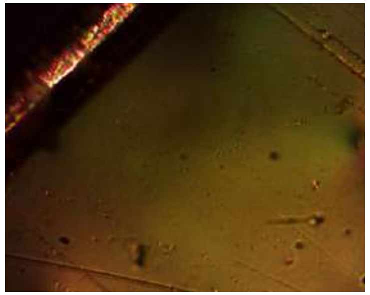
60°



70°



80°



90°

Remarks

- Although the initial results from the project looks promising especially with the distinct patterns observed at the boundaries of the 5CB covered plastic sample, a large part of the detection process is being observed through the topological difference arising from the thickness of the plastic sample, this puts into question the efficacy of the liquid crystal sample being used.
- Further research should aim to use thinner samples with measured thicknesses and observe any changes in the liquid crystal arrangement pattern.